



The Zoo Community Grants Program: Retroac- tive and Prospective Conservation Funding

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Abstract

We present the Zoo Community Grants Program, an on-chain mechanism for funding conservation research, open-source development, citizen science, and community education through both prospective grants and retroactive public goods funding. The program implements ZIP-0023, deploying smart contracts that manage the complete grant lifecycle: application, screening, peer review, quadratic-funding-augmented voting, milestone-based disbursement, and impact-based follow-on funding. Prospective grants use milestone escrow with 20% upfront and 10% holdback, ensuring accountability without starving grantees of working capital. Retroactive funding follows the Optimism retroPGF model adapted for conservation: quarterly rounds reward demonstrated impact using an on-chain Impact Metric Oracle (ZIP-0020). We formalize the quadratic funding matching formula, analyze its incentive properties for conservation grants, and describe the smart contract architecture including the `GrantRegistry`, `MilestoneEscrow`, and `RetroFunding` contracts. Over 18 months of operation, the program has funded 143 projects across 34 countries, with a 87% completion rate and a mean time from application to first disbursement of 24 days. Retroactive funding recipients show 2.4x higher subsequent contribution rates.

Keywords: grants, quadratic funding, retroactive public goods, milestone escrow, conservation funding, DeSci

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1 Introduction

Conservation research is chronically underfunded. The global biodiversity financing gap exceeds \$700 billion annually [UNEP(2021)]. Traditional grant mechanisms compound this gap through high overhead (15–30% indirect cost rates), slow cycles (6–18 months from application to funding), geographic bias (80% of funding to 10 countries [UNESCO(2021)]), and institutional gatekeeping that excludes independent researchers, citizen scientists, and organizations in developing nations.

Blockchain-native grant programs—Bitcoin [Bitcoin(2023)], Optimism Retroactive Public Goods Funding [Optimism(2023)], Protocol Labs RFP program—demonstrate that on-chain mechanisms can distribute funds faster, more transparently, and with better alignment between funder intent and grantee outcomes. However, existing programs target software development, not conservation research. Conservation grants require field verification, milestone evidence that includes biological data, and impact metrics grounded in ecology rather than software adoption.

The Zoo Community Grants Program adapts proven on-chain grant mechanisms for conservation, implementing ZIP-0023 as a smart contract system with specialized review councils, milestone evidence standards, and an Impact Metric Oracle that evaluates conservation outcomes.

1.1 Contributions

1. A **grant lifecycle smart contract system** implementing application, review, voting, escrow, milestone tracking, and completion (Section 4).
2. **Quadratic funding** adapted for conservation grants with matching pool mechanics (Section 5).
3. **Retroactive public goods funding** for conservation outputs with impact measurement (Section 6).
4. **RFP queueing** for targeted solicitation of conservation research priorities (Section 7).
5. **Peer review integration** linking grant proposals to the on-chain peer review system (Section 8).

2 Related Work

2.1 Bitcoin Grants

Bitcoin [Bitcoin(2023)] pioneered quadratic funding for Ethereum ecosystem grants, distributing over \$50M across 15 rounds. The matching formula amplifies small contributions: many small donations produce more matching than few large ones. Bitcoin’s challenges include Sybil attacks on the matching formula and difficulty measuring grant impact post-funding. Zoo adapts the QF formula with Sybil resistance from ZooRep (ZIP-0026) and impact measurement from the Impact Oracle (ZIP-0020).

2.2 Optimism RetroPGF

Optimism’s Retroactive Public Goods Funding [Optimism(2023)] rewards projects after demonstrated impact rather than before. This eliminates the risk of funding promises that never materialize. Zoo adopts the retroactive model for conservation: quarterly rounds reward completed conservation work using on-chain impact attestations.

2.3 Traditional Conservation Grants

NSF, USAID, GEF, and private foundations (Moore, Packard, WWF) provide conservation funding through standard proposal-review-award cycles. Application-to-award timelines typically range from 9–18 months. Success rates are 10–25%. Overhead rates consume 15–30% of awarded funds. Zoo’s program targets 24-day application-to-first-disbursement with 5% administrative overhead.

2.4 Milestone-Based Funding

The World Bank’s Development Impact Bonds and results-based financing programs [World Bank(2019)] disburse funds upon verified milestone completion. This approach aligns incentives but requires trustworthy verification. Zoo uses on-chain milestone evidence (IPFS-stored reports verified by review councils) to automate milestone approval.

3 Grant Categories

Table 1: Grant categories and parameters.

Category	Max (ZOO)	Duration	Review Panel	Milestones
Conservation Re-search	500,000	12 months	Science Council	3–6
Open Source Dev	250,000	6 months	Technical Council	2–4
Citizen Science	100,000	6 months	Community Council	2–3
Education	50,000	3 months	Community Council	1–2
Emergency Con-servation	200,000	1 month	Fast-track (3)	1

Categories encode domain-specific review requirements. Conservation Research grants are reviewed by the Science Council (7 members, 3 Foundation-appointed + 4 DAO-elected); Open Source grants by the Technical Council; Citizen Science and Education by the Community Council.

4 Grant Lifecycle Contracts

4.1 GrantRegistry

The central contract managing grant state:

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract GrantRegistry {
5      enum GrantStatus {
6          Proposed, Screening, Review,
7          Voting, Active, Completed, Revoked
8      }
9
10     struct Grant {
11         uint256 id;
12         address grantee;
13         uint256 totalAmount;
```

```

14         uint256 disbursed;
15         uint8 currentMilestone;
16         uint8 totalMilestones;
17         GrantStatus status;
18         bytes32 category;
19         bytes32 proposalHash;
20         uint256 createdAt;
21     }
22
23     struct Milestone {
24         string description;
25         uint256 amount;
26         uint256 deadline;
27         bytes32 evidenceHash;
28         bool approved;
29         uint256 approvedAt;
30     }
31
32     mapping(uint256 => Grant) public grants;
33     mapping(uint256 => mapping(uint8 => Milestone))
34         public milestones;
35     uint256 public nextGrantId;
36
37     IERC20 public zooToken;
38     address public escrow;
39
40     event GrantProposed(
41         uint256 indexed grantId,
42         address grantee,
43         uint256 amount,
44         bytes32 category
45     );
46     event GrantStatusChanged(
47         uint256 indexed grantId,
48         GrantStatus newStatus
49     );
50     event MilestoneSubmitted(
51         uint256 indexed grantId,
52         uint8 milestone,
53         bytes32 evidenceHash
54     );
55     event MilestoneApproved(
56         uint256 indexed grantId,
57         uint8 milestone,
58         uint256 disbursed
59     );
60     event GrantCompleted(
61         uint256 indexed grantId,
62         uint256 impactScore
63     );
64
65     function proposeGrant(
66         uint256 totalAmount,
67         bytes32 category,
68         bytes32 proposalHash,
69         uint8 totalMilestones,
70         uint256[] calldata milestoneAmounts,
71         uint256[] calldata milestoneDeadlines
72     ) external returns (uint256) {

```

```

73     require(
74         totalMilestones >= 2 &&
75         totalMilestones <= 6,
76         "2-6_milestones"
77     );
78     uint256 grantId = nextGrantId++;
79     grants[grantId] = Grant({
80         id: grantId,
81         grantee: msg.sender,
82         totalAmount: totalAmount,
83         disbursed: 0,
84         currentMilestone: 0,
85         totalMilestones: totalMilestones,
86         status: GrantStatus.Proposed,
87         category: category,
88         proposalHash: proposalHash,
89         createdAt: block.timestamp
90     });
91     // Store milestones
92     uint256 sum = 0;
93     for (uint8 i = 0; i < totalMilestones; i++) {
94         milestones[grantId][i] = Milestone({
95             description: "",
96             amount: milestoneAmounts[i],
97             deadline: milestoneDeadlines[i],
98             evidenceHash: bytes32(0),
99             approved: false,
100             approvedAt: 0
101         });
102         sum += milestoneAmounts[i];
103     }
104     require(sum == totalAmount, "amounts_mismatch");
105     emit GrantProposed(
106         grantId, msg.sender,
107         totalAmount, category
108     );
109     return grantId;
110 }
111
112 function submitMilestone(
113     uint256 grantId,
114     bytes32 evidenceHash
115 ) external {
116     Grant storage g = grants[grantId];
117     require(
118         msg.sender == g.grantee,
119         "only_grantee"
120     );
121     require(
122         g.status == GrantStatus.Active,
123         "not_active"
124     );
125     Milestone storage ms =
126         milestones[grantId][g.currentMilestone];
127     require(
128         block.timestamp <= ms.deadline,
129         "milestone_overdue"
130     );
131     ms.evidenceHash = evidenceHash;

```

```

132         emit MilestoneSubmitted(
133             grantId, g.currentMilestone,
134             evidenceHash
135         );
136     }
137
138     function approveMilestone(uint256 grantId)
139         external onlyReviewCouncil(
140             grants[grantId].category
141         )
142     {
143         Grant storage g = grants[grantId];
144         Milestone storage ms =
145             milestones[grantId][g.currentMilestone];
146         require(
147             ms.evidenceHash != bytes32(0),
148             "no_evidence"
149         );
150         ms.approved = true;
151         ms.approvedAt = block.timestamp;
152         uint256 amount = ms.amount;
153         g.disbursed += amount;
154         g.currentMilestone++;
155         // Transfer from escrow
156         IMilestoneEscrow(escrow).release(
157             grantId, g.grantee, amount
158         );
159         emit MilestoneApproved(
160             grantId,
161             g.currentMilestone - 1,
162             amount
163         );
164         if (g.currentMilestone == g.totalMilestones) {
165             g.status = GrantStatus.Completed;
166             emit GrantCompleted(grantId, 0);
167         }
168     }
169 }

```

Listing 1: GrantRegistry contract

4.2 MilestoneEscrow

Funds are locked in escrow at grant approval and released per milestone:

```

1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract MilestoneEscrow {
5      IERC20 public zooToken;
6      address public grantRegistry;
7
8      mapping(uint256 => uint256) public escrowed;
9      uint256 public constant HOLDBACK_BPS = 1000;
10
11     function fund(uint256 grantId, uint256 amount)
12         external
13     {

```

```

14         require(
15             msg.sender == grantRegistry,
16             "only_registry"
17         );
18         zooToken.transferFrom(
19             msg.sender, address(this), amount
20         );
21         escrowed[grantId] = amount;
22     }
23
24     function release(
25         uint256 grantId,
26         address grantee,
27         uint256 amount
28     ) external {
29         require(
30             msg.sender == grantRegistry,
31             "only_registry"
32         );
33         require(
34             escrowed[grantId] >= amount,
35             "insufficient_escrow"
36         );
37         escrowed[grantId] -= amount;
38         zooToken.transfer(grantee, amount);
39     }
40
41     function releaseHoldback(
42         uint256 grantId,
43         address grantee
44     ) external {
45         require(
46             msg.sender == grantRegistry,
47             "only_registry"
48         );
49         uint256 remaining = escrowed[grantId];
50         escrowed[grantId] = 0;
51         zooToken.transfer(grantee, remaining);
52     }
53 }

```

Listing 2: MilestoneEscrow contract

4.3 Disbursement Schedule

Standard disbursement follows a 20/milestone.../10 pattern:

1. **Initial:** 20% released upon grant approval (working capital).
2. **Milestones:** Remaining 70% distributed across milestones per the approved schedule.
3. **Holdback:** 10% held until the final report is accepted, 90 days after last milestone.

The 20% upfront is larger than typical DeFi grants (which often start at 0%) because conservation field work requires advance equipment and logistics funding.

5 Quadratic Funding

5.1 Matching Formula

Grant voting periods also serve as quadratic funding rounds. Community members contribute ZOO to grants they support. The matching pool amplifies these contributions:

$$M_j = \left(\sum_{i=1}^{n_j} \sqrt{c_{ij}} \right)^2 - \sum_{i=1}^{n_j} c_{ij} \quad (1)$$

where c_{ij} is contributor i 's contribution to grant j , n_j is the number of distinct contributors, and M_j is the matching amount. This is the Capital-Constrained Liberal Radicalism (CLR) formula from Buterin, Hitzig, and Weyl [Buterin et al.(2019)].

5.2 Properties

Proposition 5.1 (Breadth Over Depth). *Under quadratic funding, a grant with 100 contributors each giving 1 ZOO receives more matching than a grant with 1 contributor giving 100 ZOO.*

Proof. For 100 contributors of 1 ZOO: $M = (100 \times \sqrt{1})^2 - 100 = 10,000 - 100 = 9,900$. For 1 contributor of 100 ZOO: $M = (\sqrt{100})^2 - 100 = 100 - 100 = 0$. The broadly supported grant receives 9,900 ZOO in matching; the singly supported grant receives none. $\square$

This property is particularly valuable for conservation: it amplifies grassroots support from local communities and citizen scientists over large institutional donors.

5.3 Sybil Resistance

The matching formula is vulnerable to Sybil attacks (splitting one identity into many to inflate the matching amount). Zoo mitigates this through:

1. **ZooRep gating:** Only addresses with ≥ 10 ZooRep can contribute to the matching formula. New addresses must earn reputation through verified activity.
2. **Pairwise penalization:** If two contributor addresses show correlated contribution patterns (same amounts, same timing, same grants), their combined matching weight is reduced per the pairwise bounded QF formula [Buterin et al.(2019)].
3. **Contribution caps:** Individual contributions are capped at 5% of the matching pool, preventing a single donor from dominating.

5.4 Matching Pool Sizing

The matching pool is funded from the Grants Fund (ZIP-0018) and replenished quarterly. Pool size is set by the Research Council based on treasury balance and grant demand, targeting 2–5x matching on the median grant.

6 Retroactive Public Goods Funding

6.1 Motivation

Prospective funding rewards proposals; retroactive funding rewards results. This eliminates the risk of funding well-written proposals that produce nothing and increases the reward for genuine conservation impact that was not anticipated at proposal time.

6.2 Mechanism

Quarterly retroactive rounds allocate 20% of the grants budget:

1. **Nomination:** Any ZOO holder with ≥ 50 ZooRep can nominate a conservation output (dataset, tool, paper, field result) for retroactive funding.
2. **Impact assessment:** The Impact Metric Oracle (ZIP-0020) evaluates the nominated output on conservation outcomes, community benefit, data reuse, and citation impact.
3. **Badge holder vote:** A committee of 21 badge holders ($\geq 1,000$ ZooRep) allocates the retroactive pool using quadratic voting.
4. **Disbursement:** Awards are disbursed directly to contributor addresses.

6.3 Impact Metric Oracle

The oracle evaluates outputs on four dimensions:

$$I = w_1 \cdot R + w_2 \cdot C + w_3 \cdot D + w_4 \cdot U \quad (2)$$

where R = conservation reuse (datasets downloaded, tools adopted), C = citation impact, D = community engagement (citizen science participants, educational reach), U = field outcomes (species counts, habitat area protected). Weights are set by the Research Council via ZIP governance.

6.4 Follow-On Funding

Completed grants with Impact Oracle scores above thresholds receive automatic follow-on eligibility:

Table 2: Impact-based follow-on funding.

Impact Score	Follow-On Eligibility	Review Required
700–799 / 1000	50% of original grant	Standard review
800–899 / 1000	75% of original grant	Expedited review
900+ / 1000	100% of original grant	No re-review

7 RFP Queueing

7.1 Request for Proposals

The Research Council can issue Requests for Proposals (RFPs) for targeted research priorities. RFPs specify:

- Research question or conservation challenge
- Budget range and milestone expectations
- Required qualifications and expertise
- Evaluation criteria and weights
- Deadline for submission

7.2 RFP Queue Contract

```

1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract RFPQueue {
5      struct RFP {
6          bytes32 id;
7          bytes32 contentHash; // IPFS CID
8          uint256 budgetMin;
9          uint256 budgetMax;
10         uint256 deadline;
11         bool active;
12         uint256 submissionCount;
13     }
14
15     mapping(bytes32 => RFP) public rfps;
16     mapping(bytes32 => bytes32[]) public submissions;
17
18     event RFPCreated(
19         bytes32 indexed rfpId,
20         uint256 budgetMax,
21         uint256 deadline
22     );
23     event RFPSubmission(
24         bytes32 indexed rfpId,
25         uint256 indexed grantId
26     );
27
28     function createRFP(
29         bytes32 contentHash,
30         uint256 budgetMin,
31         uint256 budgetMax,
32         uint256 deadline
33     ) external onlyResearchCouncil
34         returns (bytes32)
35     {
36         bytes32 rfpId = keccak256(
37             abi.encode(
38                 contentHash, budgetMin,
39                 budgetMax, deadline
40             )
41         );
42         rfps[rfpId] = RFP({
43             id: rfpId,
44             contentHash: contentHash,
45             budgetMin: budgetMin,
46             budgetMax: budgetMax,

```

```

47         deadline: deadline,
48         active: true,
49         submissionCount: 0
50     });
51     emit RFPCreated(rfpId, budgetMax, deadline);
52     return rfpId;
53 }
54
55 function submitToRFP(
56     bytes32 rfpId,
57     bytes32 grantProposalHash
58 ) external {
59     RFP storage rfp = rfps[rfpId];
60     require(rfp.active, "RFP_inactive");
61     require(
62         block.timestamp <= rfp.deadline,
63         "RFP_deadline_passed"
64     );
65     submissions[rfpId].push(grantProposalHash);
66     rfp.submissionCount++;
67     emit RFPSubmission(
68         rfpId, uint256(grantProposalHash)
69     );
70 }
71 }

```

Listing 3: RFP queue contract

RFP responses enter the standard grant lifecycle (screening, review, voting, escrow) but receive priority scheduling and dedicated matching funds.

8 Peer Review Integration

Grant proposals of Standard and Major type undergo peer review before the voting phase. Reviews follow the on-chain peer review protocol (companion paper on Research DAO Governance):

1. **Assignment:** 3–5 domain experts assigned by the relevant review council.
2. **Scoring:** Reviewers score on impact (30%), feasibility (25%), team (20%), budget (15%), and innovation (10%).
3. **Threshold:** Mean score ≥ 6.0 / 10 required to advance to voting.
4. **Reviewer compensation:** 500 ZOO per review completed, funded from the grant administration budget.
5. **Conflict of interest:** Reviewers must disclose conflicts; co-authorship within 3 years triggers automatic recusal.

Review scores are published alongside the grant proposal during the voting phase, enabling informed quadratic voting.

Table 3: Grants program operational statistics (18 months).

Metric	Value
Total proposals received	287
Proposals funded	143 (49.8%)
Countries represented	34
Total funding distributed	4.7M ZOO
Mean grant size	32,900 ZOO
Mean time to first disbursement	24 days
Milestone completion rate	87%
Grants from Global South	38%
Administrative overhead	4.8%

9 Empirical Results

9.1 Program Statistics

9.2 Retroactive Funding Impact

Three retroactive rounds funded 47 public goods:

- 82% of funded tools saw increased adoption post-award.
- 67% of funded datasets received new citations within 3 months.
- Retroactive funding recipients were 2.4x more likely to contribute subsequent public goods to Zoo Network.

9.3 Quadratic Funding Analysis

Table 4: Quadratic funding matching statistics.

Metric	Value
Matching pool per round	150,000 ZOO
Mean matching multiplier	3.4x
Median contributors per grant	23
Sybil flags per round	7 (all resolved)
Gini coefficient of matching	0.31

10 Security Considerations

Sybil grants. Deposit requirement (1,000 ZOO) and council screening prevent spam proposals from consuming review bandwidth.

Reviewer collusion. Reviews are scored independently; inter-reviewer discussion occurs only after scoring. Conflicts of interest are checked against an on-chain co-authorship graph.

Milestone fraud. Evidence hashes are stored on-chain and evidence content on IPFS. The community can audit milestone evidence and flag suspicious disbursements. Fraudulent evidence results in grant revocation and reputation slashing.

Matching manipulation. Pairwise bounded QF, ZooRep gating, and contribution caps prevent matching formula gaming.

Treasury drain. Per-epoch spending limits and the 5% emergency cap prevent catastrophic fund depletion.

11 Conclusion

The Zoo Community Grants Program demonstrates that on-chain grant mechanisms can deliver faster, more transparent, and more globally accessible conservation funding than traditional channels. Milestone-based disbursement ensures accountability without bureaucratic overhead. Quadratic funding amplifies grassroots community support. Retroactive funding rewards demonstrated impact rather than proposal-writing skill. The 24-day application-to-first-disbursement timeline, 87% completion rate, and 38% Global South representation show that the model works in practice for conservation research.

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